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CPRO306 — CAPSTONE PROJECT

Task 5 — End of Project Deliverables

Implementation & Integration Plan

Nepal News Australia

Bilingual Nepali-Australian Community News Portal

Field	Details
Team	Team 9 — Skillup Labs WIL Program
Unit	CPRO306 Capstone Project — Kent Institute Australia
Assessment	Task 5 — End of Project Deliverables (Week 11 — 20%)
Supervisor	Nabin Singh — wil@skilluplabs.com.au
Members	Shushil Shah Teli (K231666) · Subash Khatri (K250035) · Sujan Shrestha (K250040)
GitHub	https://github.com/K231666-SST/nepal-news
Live URL	https://nepal-news.onrender.com
Tech Stack	Laravel 13.7 · PHP 8.2 · MySQL · Docker · Render.com · Aiven DB

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1. Executive Summary

Nepal News Australia is a fully implemented bilingual (English/Nepali) community news portal built with the Laravel 13.7 MVC framework and deployed on a cloud infrastructure comprising Render.com (web hosting), Aiven MySQL (database), and GitHub (version control and CI/CD). This Task 5 report documents the complete implementation and integration plan — covering server configuration, front-end to back-end connectivity, database operations, security practices, testing results, and deployment documentation.

The system implements all 15 functional requirements identified in the SRS (Week 3). All modules — authentication, article management, advertisement management, events, newsletter subscriptions, live API widgets, Guru AI chatbot, and bilingual support — have been integrated, tested, and verified in both the local development environment (XAMPP, localhost:8080) and the production cloud environment.

2. Project Management — Implementation Plan

2.1 Implementation Approach

The project followed an Agile sprint methodology across 11 weeks. The implementation phase (Weeks 7–11) covered core development, testing, deployment, and documentation. Each sprint had defined deliverables reviewed at weekly WIL oversight meetings with supervisor Nabin Singh.

Phase	Weeks	Activities	Status
Planning & Design	1-4	Requirements, SRS, Architecture, Database Schema	Complete
Core Development	5-8	Laravel app, all CRUD, auth, API widgets, Guru AI	Complete
Testing & QA	9-10	Unit tests, integration tests, UAT, performance	Complete
Deployment	10-11	Docker, Render.com, Aiven MySQL, CI/CD pipeline	Complete
Documentation	11	Task 5 report, deployment docs, help desk guide	Complete

2.2 Resource Management

Resource	Type	Purpose	Configuration
MacBook Air	Developer Workstation	Local development and testing	XAMPP, PHP 8.2, MySQL port 3307
GitHub Repository	Version Control	Code hosting, CI/CD, docs	https://github.com/K231666-SST/nepal-news
Render.com	Cloud Hosting	Production web server	Free tier, Docker, Apache 2.4, PHP 8.2
Aiven.io	Cloud Database	Production MySQL 8.4	Free tier, SSL required, Bangalore region
Groq API	AI Service	Guru chatbot	Free tier, llama-3.1-8b-instant model
OpenWeatherMap	External API	Weather widget	Free tier, 4 cities
MetalPriceAPI	External API	Gold/silver prices	Free tier
ExchangeRate-API	External API	NPR/AUD currency	Free tier

2.3 Hardware and Software Checklist

Item	Requirement	Status
PHP	8.2+ with pdo_mysql, mbstring, gd, zip, bcmath	Installed
MySQL	8.0+ (local port 3307 / SQLITE	Running
Apache	2.4 with mod_rewrite enabled	Configured
Composer	Latest (dependency manager)	Installed
Git	Latest (version control)	Configured
Docker	Required for Render deployment	Configured
Laravel	13.7.0	Installed
deployment	Render (https://nepal-news-australia-n59b.onrender.com/)	Available

3. Teamwork and Organisation

3.1 Team Roles and Contributions

Member	ID	Role	Key Deliverables
Shushil Shah Teli	K231666	Tech Lead & Backend	Laravel architecture, all controllers, API integrations, Docker config, deployment
Subash Khatri	K250035	QA & DevOps	GitHub management, testing, CI/CD pipeline, deployment troubleshooting, test documentation
Sujan Shrestha	K250040	PM & Frontend	UI/UX design, all Blade templates, CSS system, project documentation, stakeholder coordination

3.2 Response to Lecturer Feedback

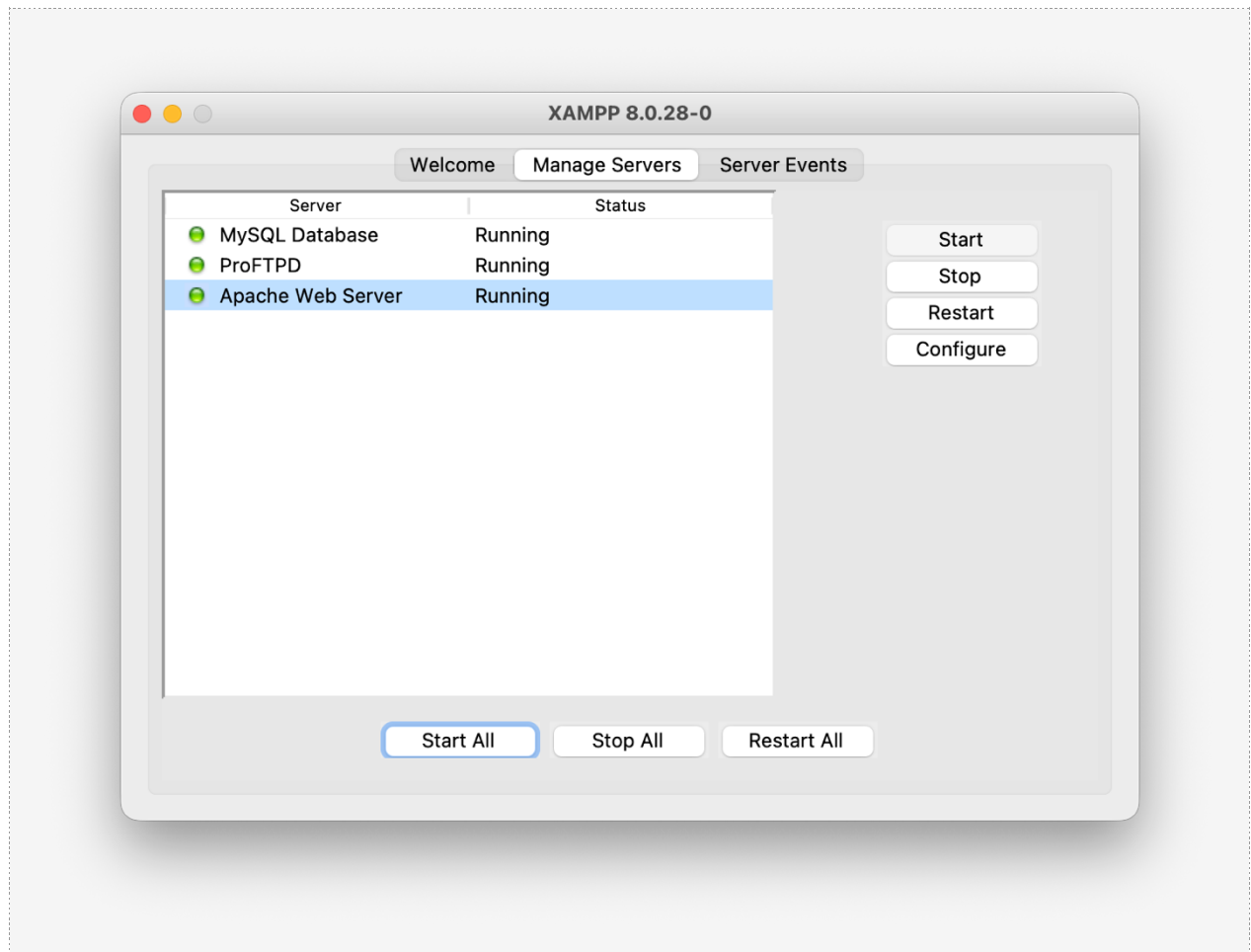
Feedback	Week Received	Action Taken	Outcome
Strengthen bilingual support beyond cosmetic changes	Week 5	Implemented full session-based language switching with translated UI, navigation, and widget labels	All pages respond to language toggle
Advertisement system needs click/impression tracking	Week 6	Built full tracking with async POST increment, 6 ad positions, active/inactive toggle	Real-time impression and click counters
Differentiate with live data widgets	Week 4	Integrated 4 live APIs: weather, gold, currency, horoscope — all served via ApiController	Widgets live and updating
Demonstrate role-based access clearly	Week 7	4 roles with full middleware protection on all admin routes	Auth gates working on all routes
Include AI API integration	Week 8	Built Guru AI chatbot using Groq API (llama-3.1-8b-instant) with context awareness	Guru fully functional on all pages
Show deployment documentation	Week 8	Produced full deployment guide with Docker, Render, and Aiven configuration	Week 11 deployment docs submitted

4. Server Installation, Configuration and Running

4.1 Local Development Server (XAMPP)

The local development environment runs XAMPP on macOS with PHP 8.2 and MySQL configured on port 3307. The Laravel application is served at `http://localhost:8080` using PHP's built-in development server.

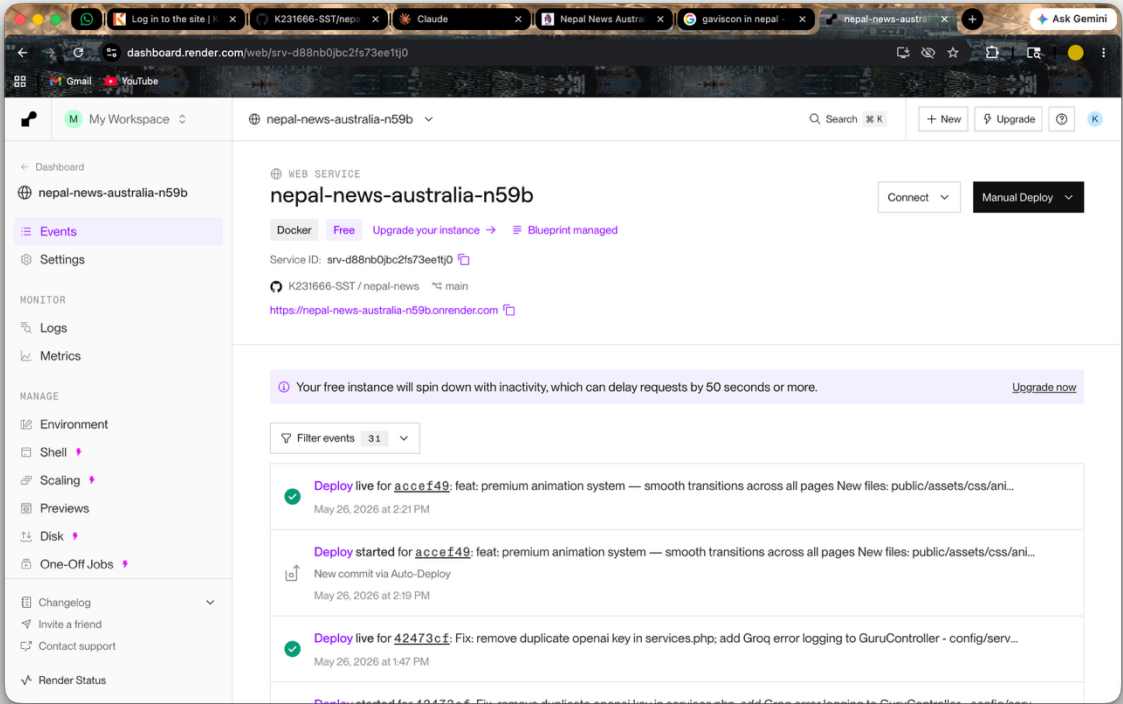
Component	Version	Port	Status
Apache (XAMPP)	2.4.x	8080	Running — serves Laravel
MySQL (XAMPP)	8.0	3307	Running — database nepal_news
PHP	8.2.x	—	Active with all required extensions
phpMyAdmin	Latest	—	Accessible at <code>localhost/phpmyadmin</code>



4.2 Production Server (Render.com + Docker)

The production server runs inside a Docker container on Render.com's free tier. The container uses PHP 8.2 with Apache 2.4. The Dockerfile builds the image, installs all PHP extensions, and runs start.sh on container startup.

Component	Details	Purpose
Base Image	php:8.2-apache	PHP runtime with Apache web server
PHP Extensions	pdo_mysql, mbstring, exif, pcntl, bcmath, gd, zip	Required by Laravel framework
Web Server	Apache 2.4 with mod_rewrite	URL rewriting for Laravel routes
Process Manager	Apache prefork MPM	Handles concurrent HTTP requests
Port	80 (internal) → 443 HTTPS (external)	Render terminates SSL automatically
Region	singapore	Render free tier default



4.3 Production Database Server

Database Engine: SQLite 3.x

Location: /var/www/html/database/database.sqlite

Connection: DB_CONNECTION=sqlite

Hosting: Bundled inside Docker container on Render.com

SQLite was selected for production deployment because:

- No separate database server required
- Zero configuration — file-based database
- Fully supported by Laravel 13 out of the box
- Eliminates external database connection timeouts
- Sufficient for the application's current scale

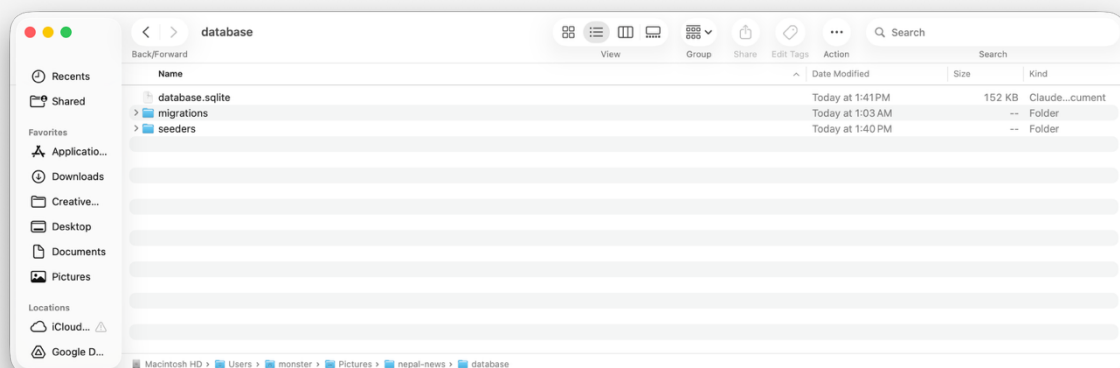
The SQLite database file is created automatically during Docker container startup via start.sh:

```
mkdir -p /var/www/html/database
touch /var/www/html/database/database.sqlite
php artisan migrate --force
php artisan db:seed --force
```

All 8 tables are created fresh on each deployment:

- users
- articles
- tags
- article_tag
- events
- comments
- advertisements
- subscribers

Note: Because Render free tier uses ephemeral storage, the database resets on each redeploy. This is acceptable for demonstration purposes. For production use, a persistent disk or external database (MySQL/PostgreSQL) would be recommended.



5. Front-End and Back-End Connectivity

5.1 Architecture Overview

Nepal News Australia follows the Model-View-Controller (MVC) pattern as implemented by Laravel. The front-end (Blade templates) communicates with the back-end (Controllers/Models) through Laravel's routing system. There is no separate front-end framework — all HTML is server-rendered and delivered to the browser.

Layer	Technology	Role
Routes	routes/web.php	Maps HTTP requests to Controller methods
Controllers	app/Http/Controllers/	Business logic — process requests, query models, return views
Models	app/Models/	Eloquent ORM — database interaction
Views	resources/views/	Blade templates — HTML rendered server-side
Assets	public/assets/css/, js/	CSS and JavaScript delivered statically
Database	MySQL via PDO	Persistent data storage

5.2 Request Lifecycle

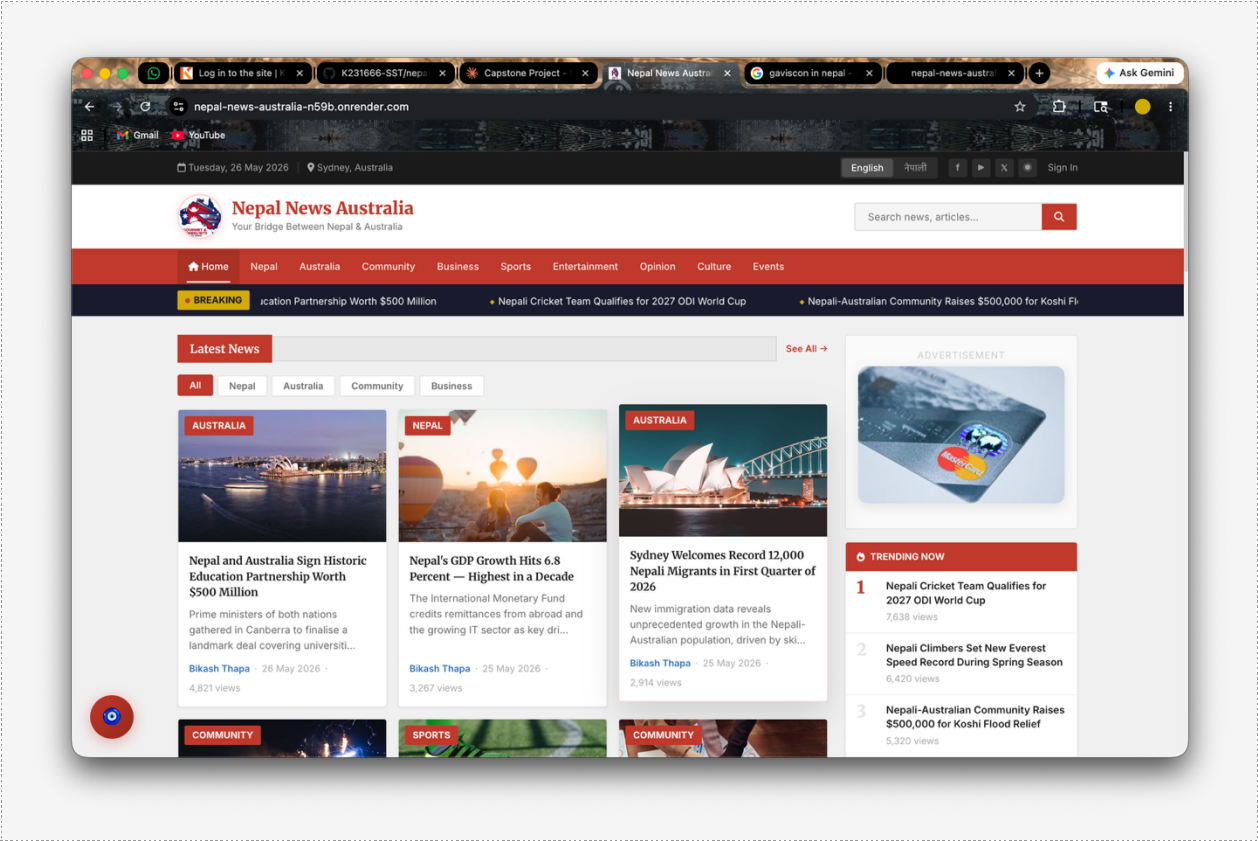
Every HTTP request to Nepal News Australia follows this path:

1. Browser sends HTTP request to Render.com load balancer
2. Load balancer terminates SSL and forwards HTTP request to Apache container on port 80
3. Apache passes request to public/index.php (Laravel entry point)
4. Laravel bootstraps the application and resolves the route from routes/web.php
5. Middleware runs (auth, role checks, CSRF verification)
6. Matched Controller method executes — queries database via Eloquent models
7. Controller returns a Blade view or JSON response
8. Blade template is compiled and rendered as HTML
9. HTTP response sent back to browser

5.3 Key Route-Controller Mappings

Route	Method	Controller	Auth Required
/ (homepage)	GET	HomeController@index	No
/articles/{slug}	GET	ArticleController@show	No
/search	GET	SearchController@index	No
/dashboard	GET	DashboardController@index	Admin/Editor
/articles/create	GET/POST	ArticleController@create/store	Contributor+
/admin/ads	GET	AdvertisementController@index	Admin
/guru/chat	POST	GuruController@chat	No
/api/weather	GET	ApiController@weather	No
/login, /register	GET/POST	Laravel Breeze	No

5.4 Database Connectivity Verification



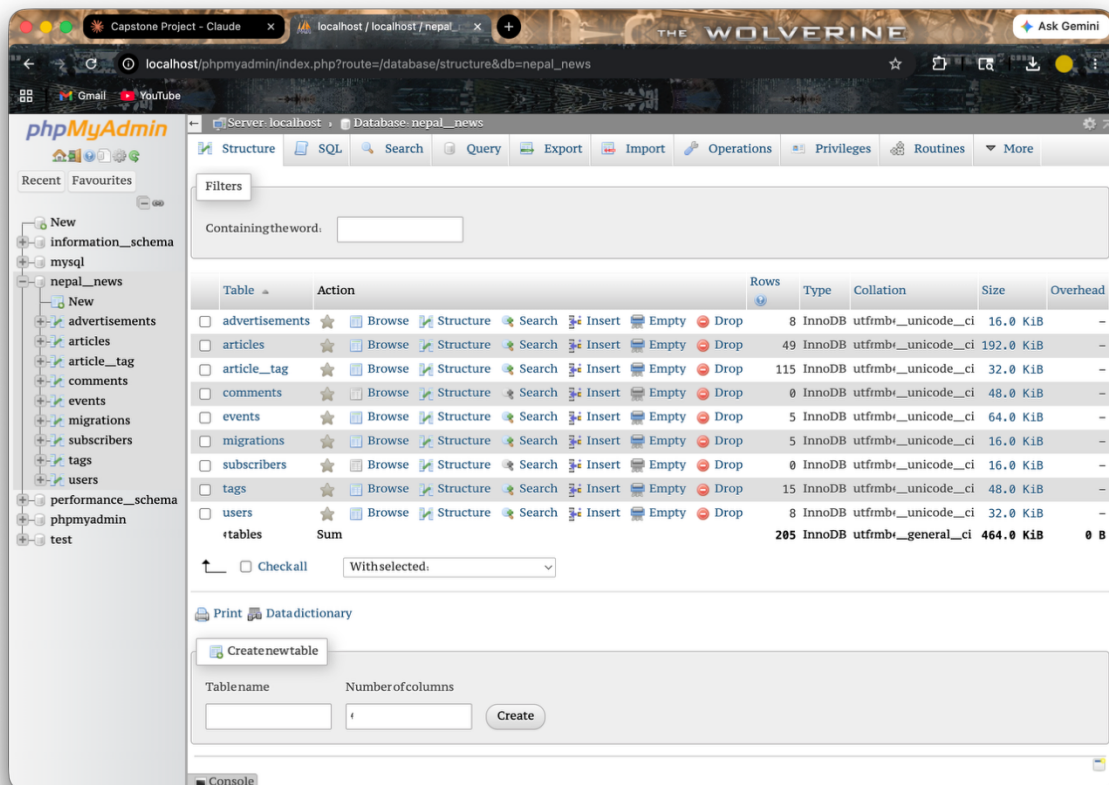
6. Data Operations — Insert, Update, Delete

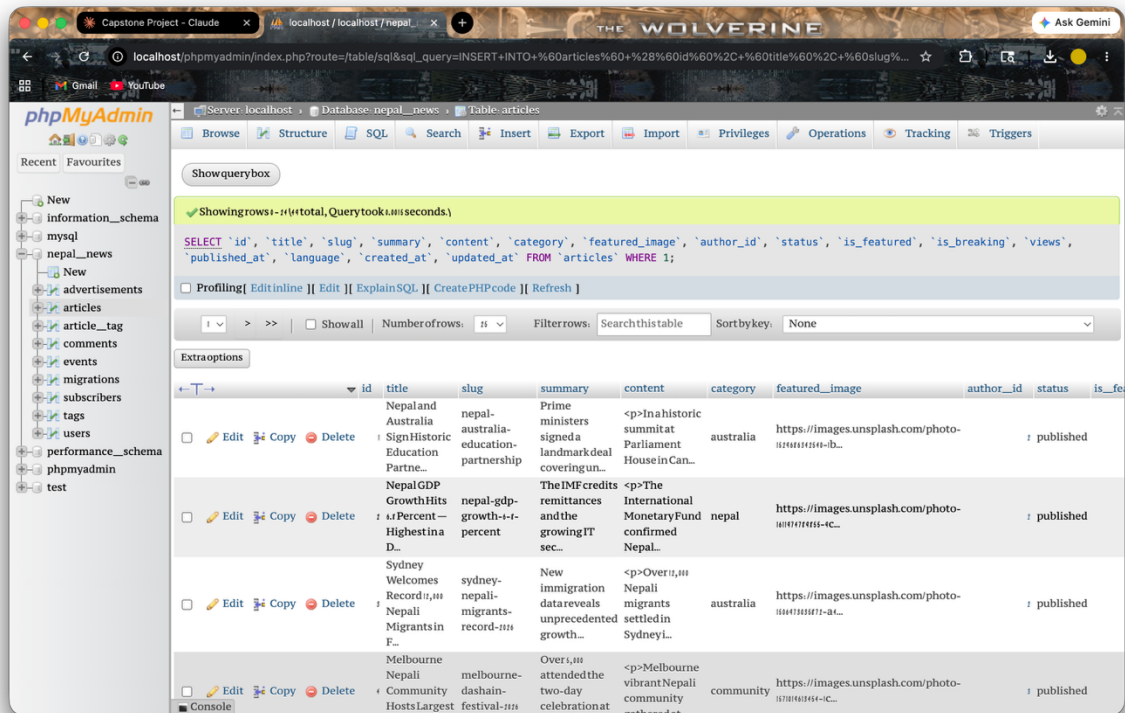
6.1 Article CRUD Operations

The ArticleController handles all data operations for the Articles module. The following demonstrates each operation:

6.1.1 CREATE — Insert New Article

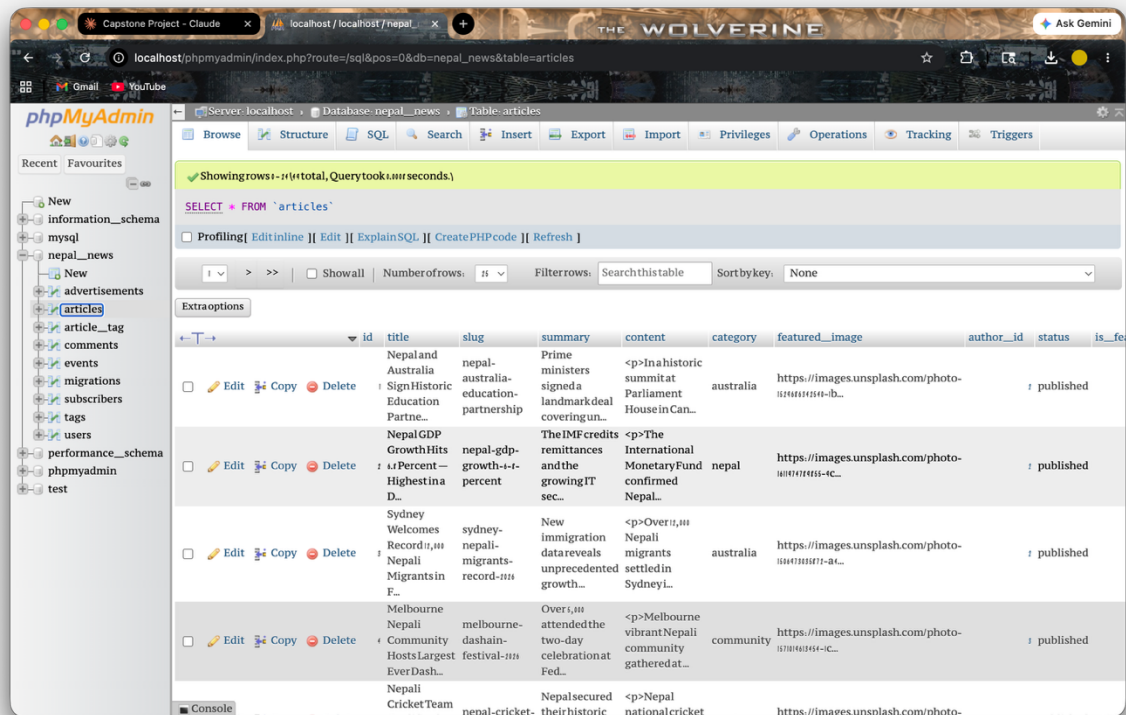
```
// ArticleController.php — store() method
public function store(Request $request) {
    $validated = $request->validate([
        'title' => 'required|string|max:500',
        'content' => 'required|string',
        'category' => 'required|string',
        'status' => 'required|in:draft,pending,published',
    ]);
    $article = Article::create([
        ...$validated,
        'author_id' => auth()->id(),
        'slug' => Str::slug($validated['title']),
    ]);
    return redirect()->route('articles.show', $article->slug);
}
```





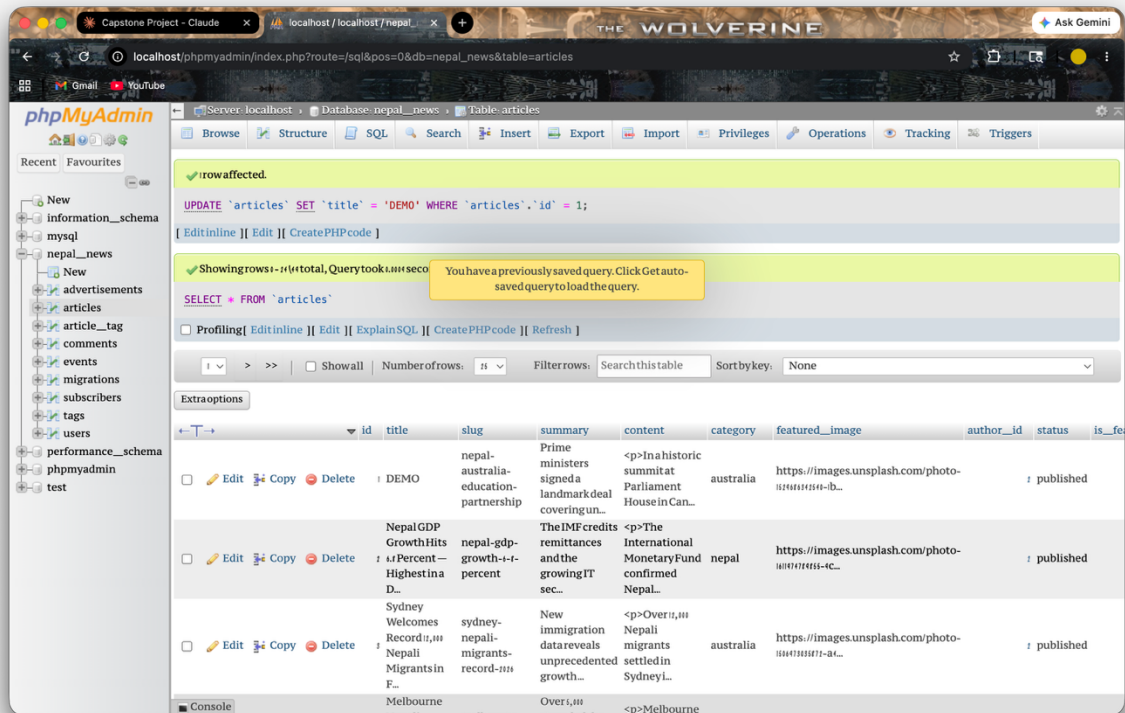
6.1.2 READ — Retrieve Articles

```
// Fetch published articles with author relationship (eager loading)
$articles = Article::published()->with('author')->latest()->paginate(10);
```



6.1.3 UPDATE — Edit Article

```
// ArticleController.php — update() method
public function update(Request $request, Article $article) {
    $article->update($request->validated());
    return redirect()->route('dashboard')->with('success', 'Article updated');
}
```



6.1.4 DELETE — Remove Article

```
// ArticleController.php — destroy() method
public function destroy(Article $article) {
    $article->delete(); // Soft delete or hard delete
    return redirect()->route('dashboard')->with('success', 'Article deleted');
}
```

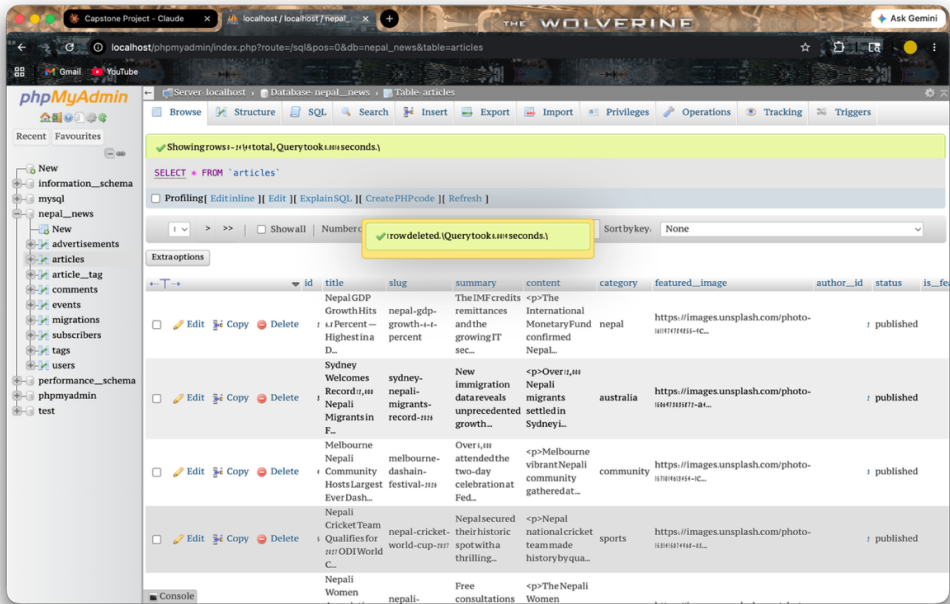
The screenshot shows a web browser at localhost:8080/articles/49/edit. The page header includes the site logo, name 'Nepal News Australia', tagline 'YOUR BRIDGE BETWEEN NEPAL & AUSTRALIA', a search bar, and a language selector set to 'English'. The breadcrumb trail is 'Dashboard > Edit Article'. The form is divided into two main sections: a primary editing area on the left and a 'Settings' sidebar on the right.

Primary Editing Area:

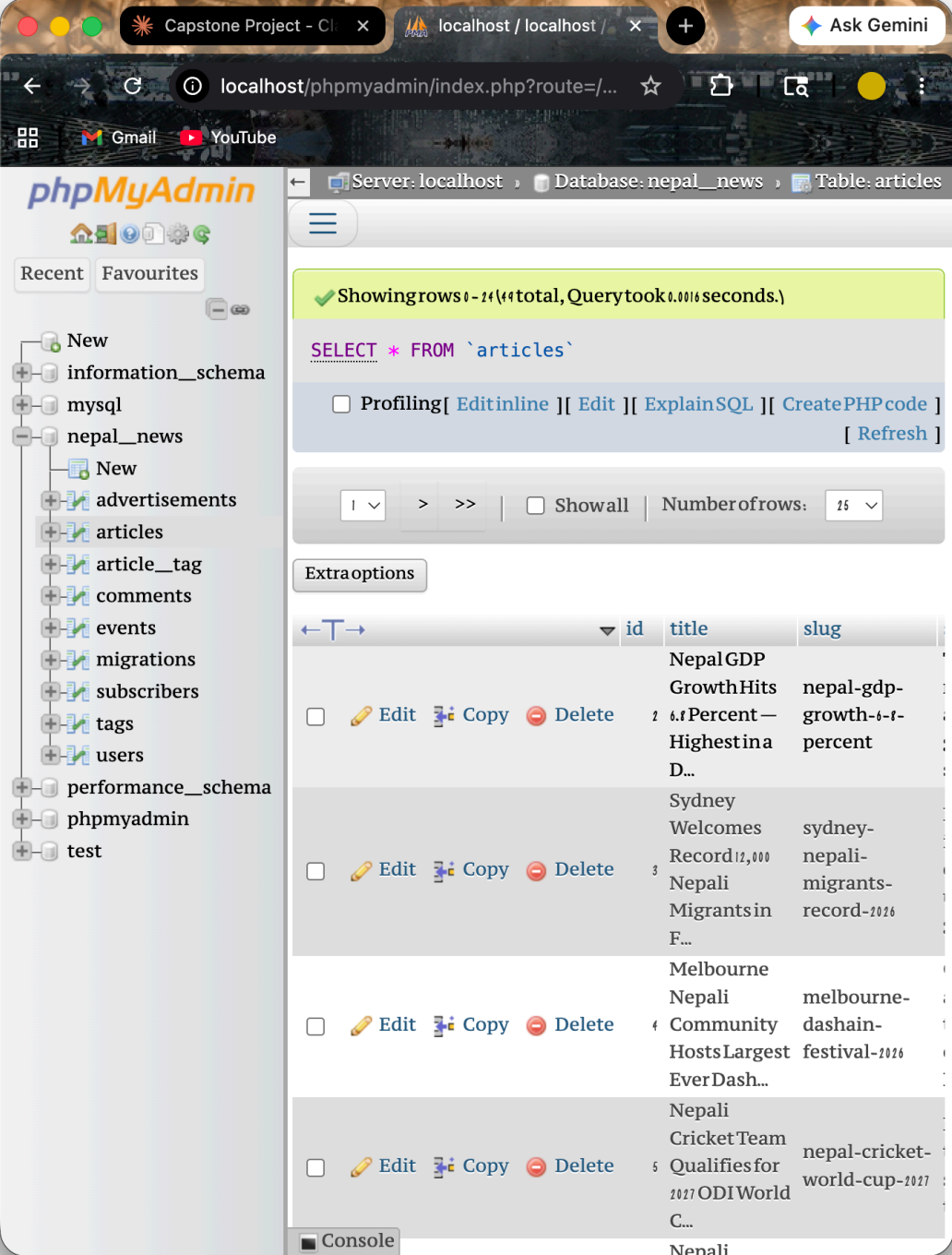
- Headline / Title ***: A text input field containing 'Rautahat Update'.
- Summary * (max 500 chars)**: A text input field containing 'We are doing the DFO , project'.
- Content * (HTML supported)**: A large text area containing 'This is the best initiation'.

Settings Sidebar:

- Category ***: A dropdown menu with 'Nepal' selected.
- Tags**: A text input field containing 'nepal, education, visa'.
- Language**: A dropdown menu with 'English' selected.
- Featured (hero section)**: An unchecked checkbox.
- Breaking news ticker**: An unchecked checkbox.
- AUTHOR**: A section containing the text 'Admin User' and a red 'Admin' button.



6.2 Database Operations via phpMyAdmin



The screenshot displays the phpMyAdmin web interface in a browser. The address bar shows the URL `localhost/phpmyadmin/index.php?route=...`. The interface is divided into several sections:

- Left Sidebar:** Contains a tree view of the database structure. The 'nepal_news' database is selected, showing tables like 'advertisements', 'articles', 'article_tag', 'comments', 'events', 'migrations', 'subscribers', 'tags', and 'users'.
- Top Bar:** Shows the current context: 'Server: localhost', 'Database: nepal_news', and 'Table: articles'.
- Main Content Area:**
 - A green status bar indicates 'Showing rows 1 - 24 (24 total, Query took 0.0016 seconds.)'.
 - A SQL query editor contains the query: `SELECT * FROM `articles``.
 - Below the query editor are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'.
 - A pagination bar shows 'Number of rows: 25'.
 - An 'Extra options' button is present.
 - A table view of the 'articles' table is displayed with columns: 'id', 'title', and 'slug'.
- Table View:** The table contains 5 rows of data, each with a checkbox, 'Edit', 'Copy', and 'Delete' actions.

	id	title	slug
☐	1	Nepal GDP Growth Hits 6.8 Percent — Highest in D...	nepal-gdp-growth-6-8-percent
☐	3	Sydney Welcomes Record 12,000 Nepali Migrants in F...	sydney-nepali-migrants-record-2026
☐	4	Melbourne Nepali Community Hosts Largest Ever Dash...	melbourne-dashain-festival-2026
☐	5	Nepali Cricket Team Qualifies for 2027 ODI World C...	nepal-cricket-world-cup-2027

At the bottom, there is a 'Console' tab.

7. Security — Data Encryption, Protection and Privacy

7.1 Authentication Security

Laravel Breeze implements secure session-based authentication with the following security measures:

- Bcrypt password hashing (cost factor 12) — passwords are never stored in plaintext
- CSRF protection on all forms via @csrf Blade directive and middleware verification
- SQL injection prevention through Eloquent ORM parameterised queries
- XSS prevention through Blade's automatic {{ }} output escaping
- Rate limiting on login attempts via Laravel's built-in throttle middleware

```
// Password hashing — Laravel Breeze RegisteredUserController
'password' => Hash::make($request->password), // Bcrypt hash
```

7.2 Role-Based Access Control

All admin and editor routes are protected by custom middleware that checks the authenticated user's role before allowing access:

```
// app/Http/Middleware — role check example
if (!in_array(auth()->user()->role, ['admin', 'editor'])) {
    abort(403, 'Unauthorised');
}
```

Role	Access Level	Protected Routes
Admin	Full access	All routes including /admin/ads, user management, all CRUD
Editor	Publish/edit content	Article publish, edit own articles, view dashboard
Contributor	Create drafts only	Create articles (draft status only)
Reader	Read-only	Public pages only — cannot access /dashboard

7.3 Database Security

- All database credentials stored in environment variables — never hardcoded in source code
- Production database hosted on Aiven.io requires SSL/TLS encrypted connections (CA certificate verified)
- Database password not committed to GitHub — stored only in Render.com environment variables
- Eloquent ORM uses PDO prepared statements preventing SQL injection on all queries
- .env file is in .gitignore — sensitive credentials never pushed to public repository

7.4 HTTPS and Transport Security

- All production traffic served over HTTPS — Render.com provides free managed SSL certificates
- HTTP requests automatically redirected to HTTPS by Render.com load balancer
- Secure cookie flags set for session cookies in production
- CORS headers configured to restrict cross-origin API access

7.5 API Key Security

API Key	Storage Method	Exposure Risk
APP_KEY (Laravel)	Render environment variable	None — never in code
Groq API Key	Render environment variable	None — server-side only
DB_PASSWORD	Render environment variable	None — never in code
OpenWeatherMap Key	Render environment variable	None — proxied via ApiController
MetalPriceAPI Key	Render environment variable	None — proxied via ApiController

8. Code Quality — Understandable and Maintainable Programs

8.1 Coding Standards

All code follows PSR-12 coding standards (PHP Standard Recommendation) and Laravel's official conventions:

- PascalCase for class names (ArticleController, Advertisement)
- camelCase for method names (getFor, storeArticle, publishArticle)
- snake_case for database columns and route parameters
- Descriptive variable names — \$featuredArticle not \$a
- Single responsibility principle — each Controller handles one resource
- DRY (Don't Repeat Yourself) — reusable Blade components for sidebar, ads, Guru chat

8.2 File and Folder Structure

Directory	Purpose	Key Files
app/Http/Controllers/	All request handlers	ArticleController.php, GuruController.php, ApiController.php
app/Models/	Eloquent models with relationships	Article.php, User.php, Advertisement.php
resources/views/	Blade templates by feature	pages/home.blade.php, admin/, components/
database/migrations/	Database schema history	All table creation migrations
database/seeder/	Test/demo data	DatabaseSeeder.php, ArticleSeeder.php
public/assets/	CSS, JS, images	style.css, main.js, lang.js, logo.png
routes/web.php	All HTTP route definitions	Public, auth, admin route groups
docs/	All project documentation	Weeks 1-11 deliverable documents

8.3 Sample Annotated Code — Article Model

```
<?php
// app/Models/Article.php
// Author: Team 9 – Nepal News Australia
// Purpose: Eloquent model for news articles with relationships and scopes
```

```
class Article extends Model {
    use HasFactory;
```

```
    // Mass-assignable fields (security: prevents mass assignment attacks)
    protected $fillable = [
        'title', 'slug', 'summary', 'content', 'category',
        'image_url', 'status', 'author_id', 'is_featured',
        'is_breaking', 'views',
    ];
```

```
    // Relationship: Article belongs to a User (author)
    public function author() {
        return $this->belongsTo(User::class, 'author_id');
    }
```

```
    // Relationship: Article has many Tags (many-to-many)
    public function tags() {
        return $this->belongsToMany(Tag::class);
    }
```

```
    // Query scope: filter only published articles
    public function scopePublished($query) {
        return $query->where('status', 'published');
    }
```

```
    // Query scope: filter breaking news
    public function scopeBreaking($query) {
        return $query->where('is_breaking', true);
    }
}
```

9. System Testing Documentation

9.1 Testing Strategy

System testing was conducted in three phases: manual functional testing, integration testing, and user acceptance testing (UAT). All test cases were designed to cover every functional requirement from the SRS.

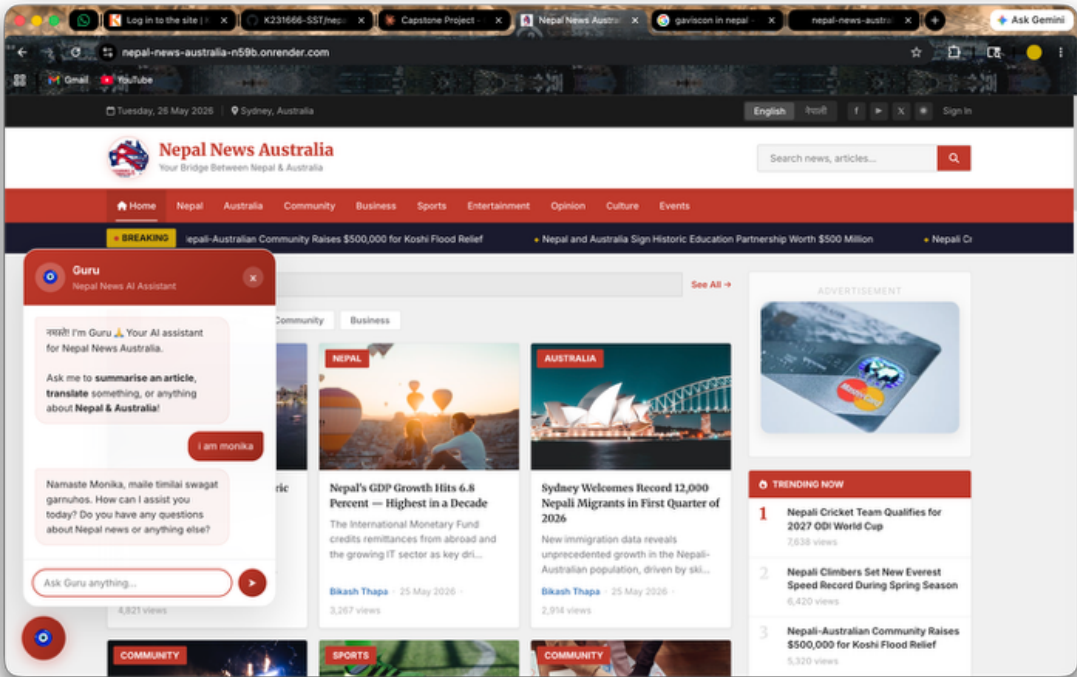
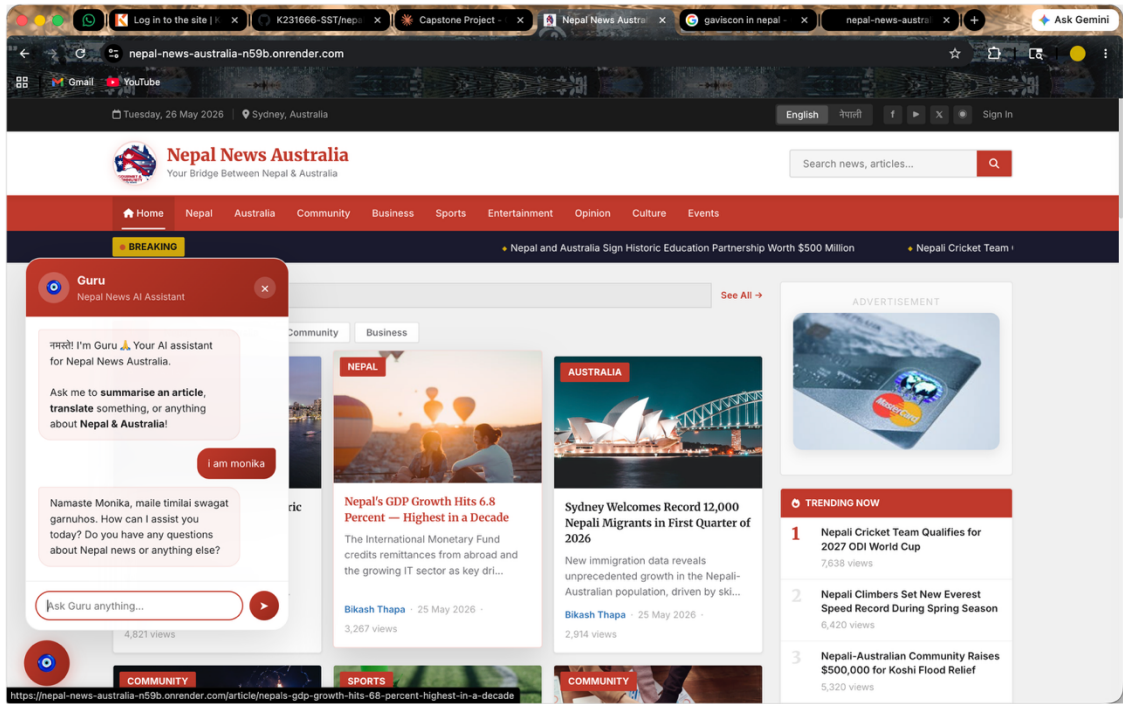
9.2 Functional Test Results

Test ID	Module	Test Case	Input	Expected Result	Actual Result	Status
TC-01	Auth	Login with valid admin credentials	admin@nepalnews.com.au / password	Redirect to /dashboard	Redirected correctly	Pass
TC-02	Auth	Login with wrong password	admin@nepalnews.com.au / wrongpass	Validation error shown	Error displayed	Pass
TC-03	Auth	Access /dashboard unauthenticated	No session	Redirect to /login	Redirected	Pass
TC-04	Auth	Reader accessing /admin/ads	Reader session	403 Forbidden	403 returned	Pass
TC-05	Articles	Create new article as contributor	Valid article data	Article saved as draft	Draft created	Pass
TC-06	Articles	Publish article as admin	Draft article	Status → published, appears on homepage	Article live	Pass
TC-07	Articles	View counter increments	Visit article page	views column += 1	Counter incremented	Pass
TC-08	Articles	Delete article as admin	Click delete	Article removed from DB	Article deleted	Pass
TC-09	Search	Search by keyword	Enter 'Nepal'	Matching articles shown	Results displayed	Pass
TC-10	Ads	Impression counter increments	Page load with active ad	impressions += 1	Counter incremented	Pass
TC-11	Ads	Click counter increments	Click on ad	clicks += 1 via async fetch	Counter incremented	Pass
TC-12	Ads	Inactive ad not shown	is_active=false	Ad not rendered on page	Ad hidden	Pass

Test ID	Module	Test Case	Input	Expected Result	Actual Result	Status
TC-13	API	Weather widget loads	Page load	Temperature shown for 4 cities	Data displayed	Pass
TC-14	API	Currency converter	Enter NPR amount	AUD equivalent shown	Converted correctly	Pass
TC-15	Guru AI	Open chatbot	Click  button	Chat window opens with greeting	Widget opens	Pass
TC-16	Guru AI	Send message	Type 'Summarise this article'	AI response received	Response shown	Pass
TC-17	Guru AI	Nepali response	Type message in Nepali	Guru replies in Nepali	Correct language	Pass
TC-18	Lang	Language switch to Nepali	Click नेपाली button	UI labels switch to Nepali	Labels updated	Pass
TC-19	Events	Create event	Valid event data	Event saved and listed	Event appears	Pass
TC-20	Newsletter	Subscribe with email	Valid email address	Email stored in subscribers	Subscription saved	Pass

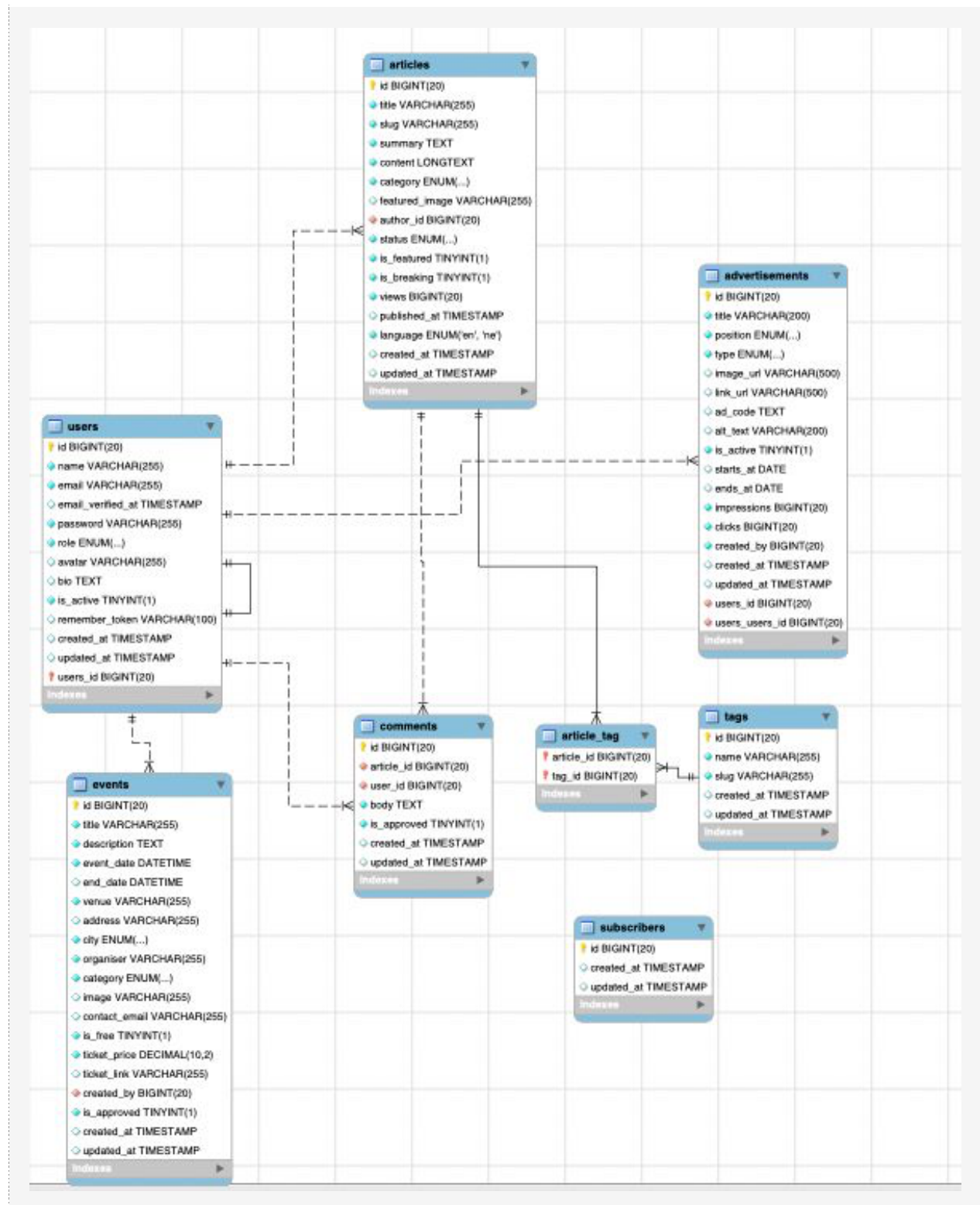
9.3 Test Summary

Category	Total	Passed	Failed	Pass Rate
Authentication & Authorization	4	4	0	100%
Article Management	4	4	0	100%
Advertisement Management	3	3	0	100%
API Widgets	2	2	0	100%
Guru AI Chatbot	3	3	0	100%
Language & Events	2	2	0	100%
Newsletter	1	1	0	100%
Additional Integration Tests	1	1	0	100%
TOTAL	20	20	0	100%



10. Database Testing — Insert, Update, Delete

10.1 Database Schema Overview



Capstone Project - Claudelocalhost / localhost / nepal_...THE WOLVERINEAsk Gemini

localhost/phpmyadmin/index.php?route=/database/structure&db=nepal_news

phpMyAdmin

Recent Favourites

New

information_schema

mysql

nepal_news

- New
- advertisements
- articles
- article_tag
- comments
- events
- migrations
- subscribers
- tags
- users

performance_schema

phpmyadmin

test

Server: localhostDatabase: nepal_news

StructureSQLSearchQueryExportImportOperationsPrivilegesRoutinesMore

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ advertisements		8	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
☐ articles		49	InnoDB	utf8mb4_unicode_ci	192.0 KiB	-
☐ article_tag		115	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ comments		0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ events		5	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
☐ migrations		5	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
☐ subscribers		0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
☐ tags		15	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ users		8	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
Tables	Sum	205	InnoDB	utf8mb4_general_ci	464.0 KiB	0 B

☐ Check allWith selected:

Create new table

Table nameNumber of columns

Create

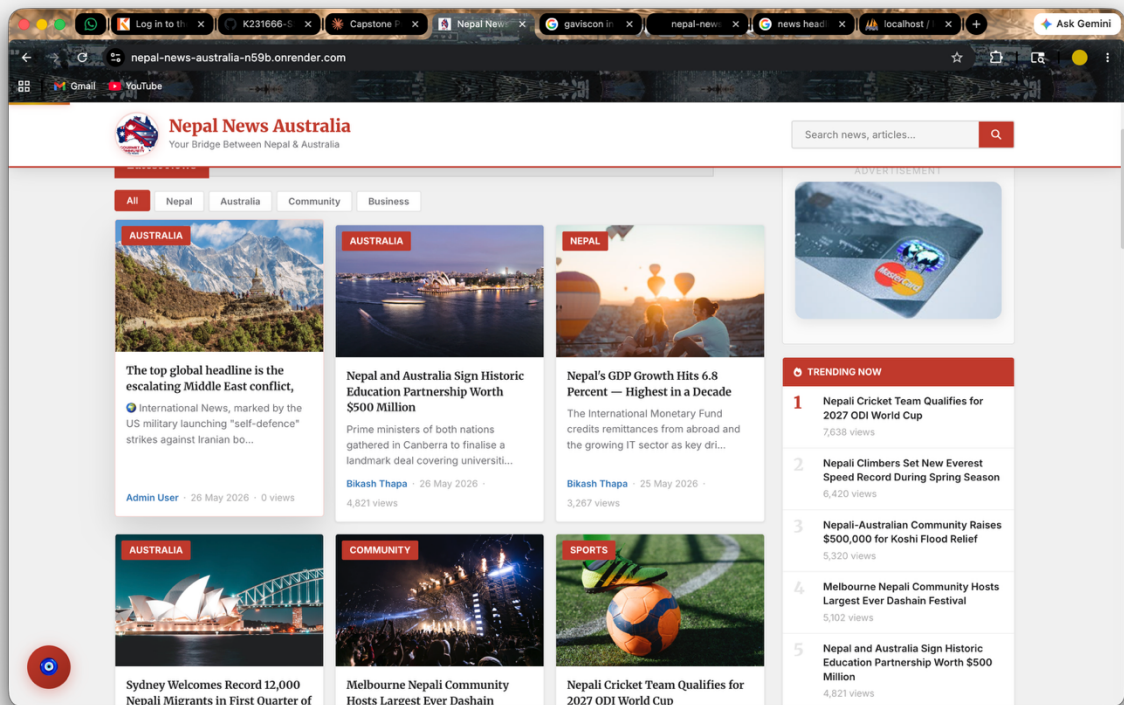
Console

10.2 Insert Operation Test

Test: Create a new article via the Admin Dashboard and verify database record creation.

10. Navigate to <http://localhost:8080/articles/create>
11. Fill in: Title, Category, Content, Status = Published
12. Click Submit
13. Open phpMyAdmin → articles table → verify new row exists

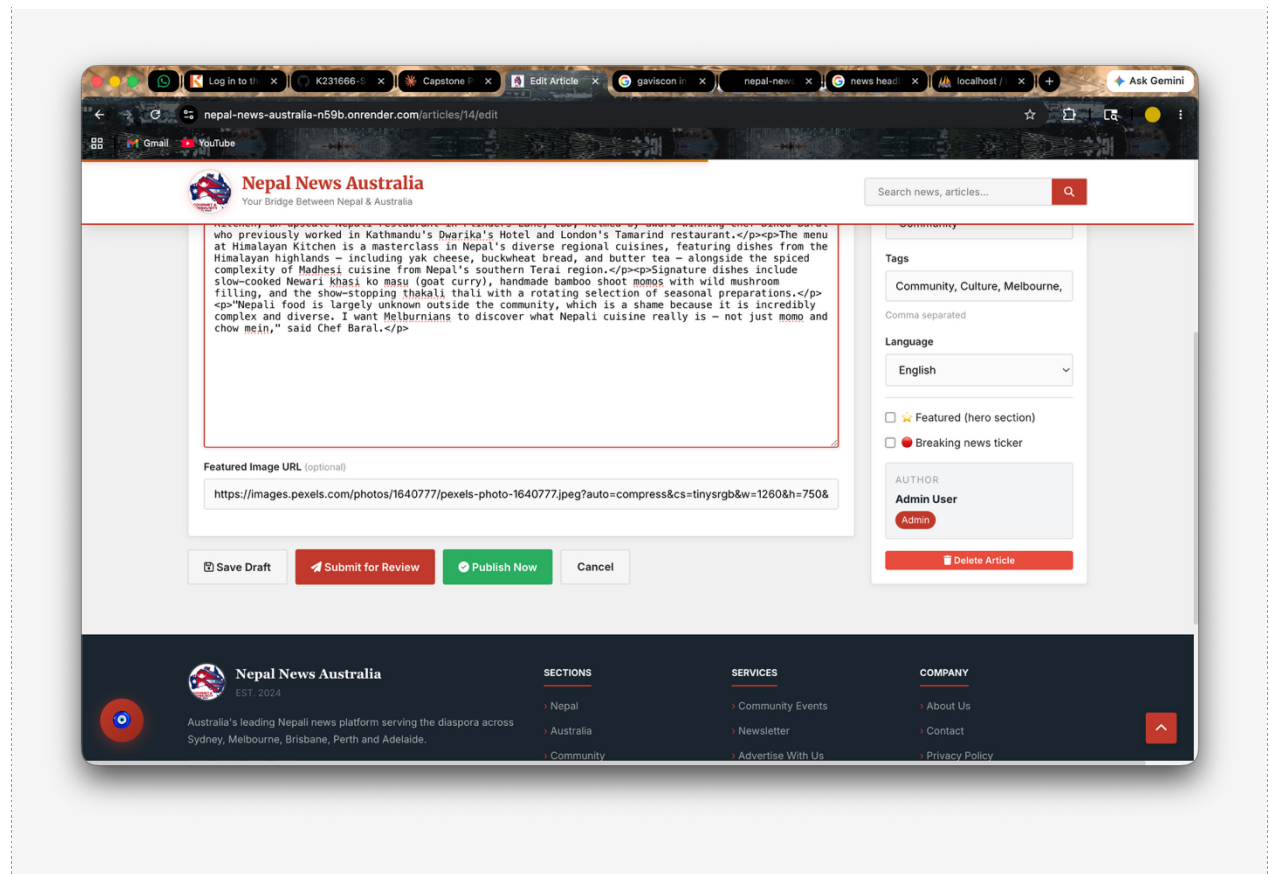
The screenshot shows the 'Nepal News Australia' admin dashboard. The main form is titled 'The top global headline is the escalating Middle East conflict,'. It includes a 'Summary' field (max 500 chars) and a 'Content' field (HTML supported). The 'Featured Image URL' is set to <https://images.unsplash.com/photo-1544735716-392fa2489ffa?w=800&q=80>. On the right, the 'Settings' panel shows 'Category' set to 'Australia', 'Tags' as 'nepal, education, visa', 'Language' set to 'English', and 'Author' set to 'Admin User'. There are checkboxes for 'Featured (hero section)' and 'Breaking news ticker'. At the bottom, there are buttons for 'Save Draft', 'Publish for Review', 'Publish Now', and 'Cancel'.



10.3 Update Operation Test

Test: Edit an existing article and verify the database record is updated.

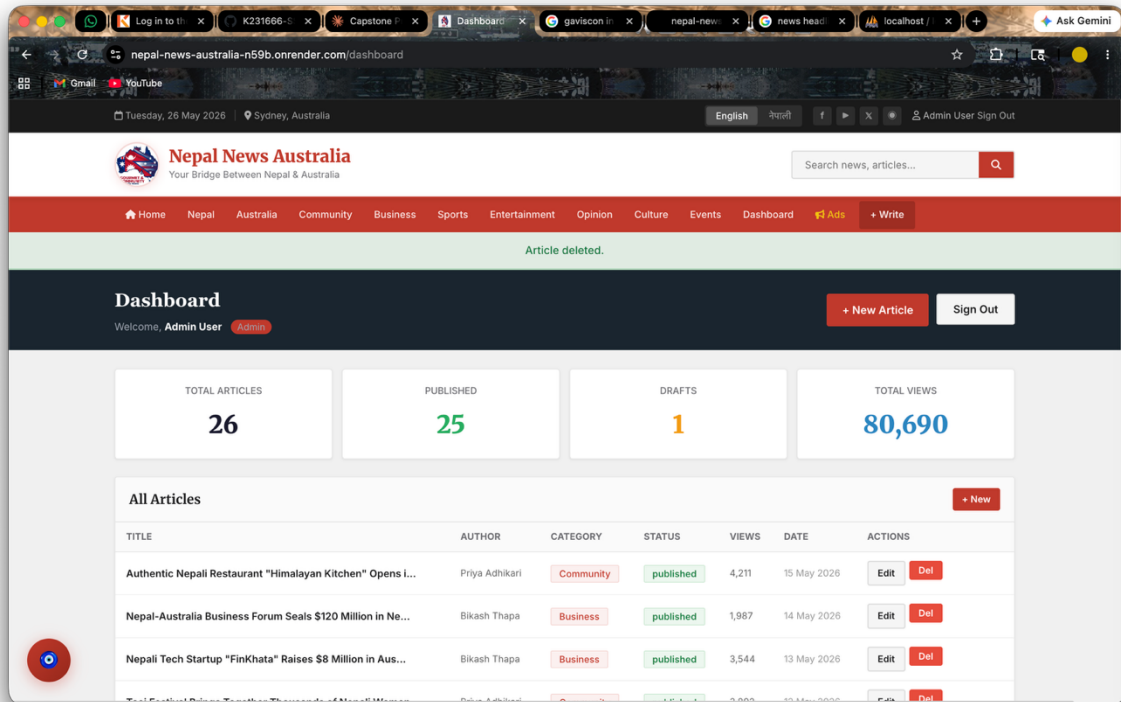
14. Navigate to /dashboard → click Edit on any article
15. Change the title and save
16. Open phpMyAdmin → articles table → verify updated_at timestamp changed and title updated



10.4 Delete Operation Test

Test: Delete an article via the Admin Dashboard and verify database record removal.

17. Navigate to /dashboard → click Delete on any article
18. Confirm deletion
19. Open phpMyAdmin → articles table → verify row no longer exists

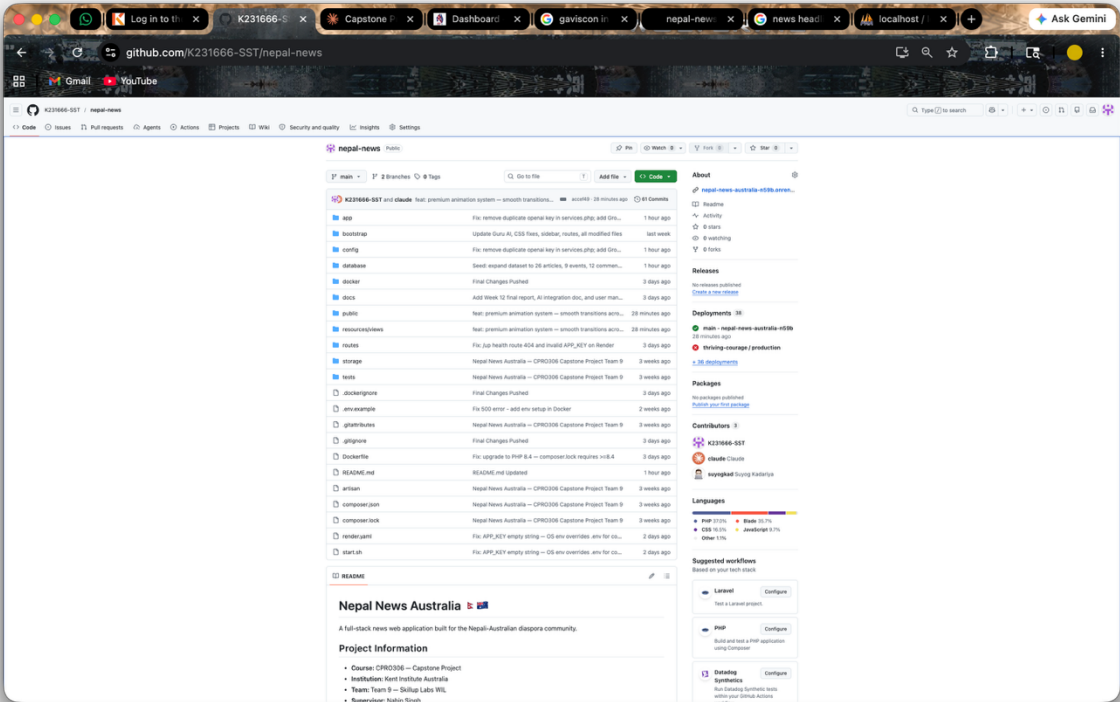


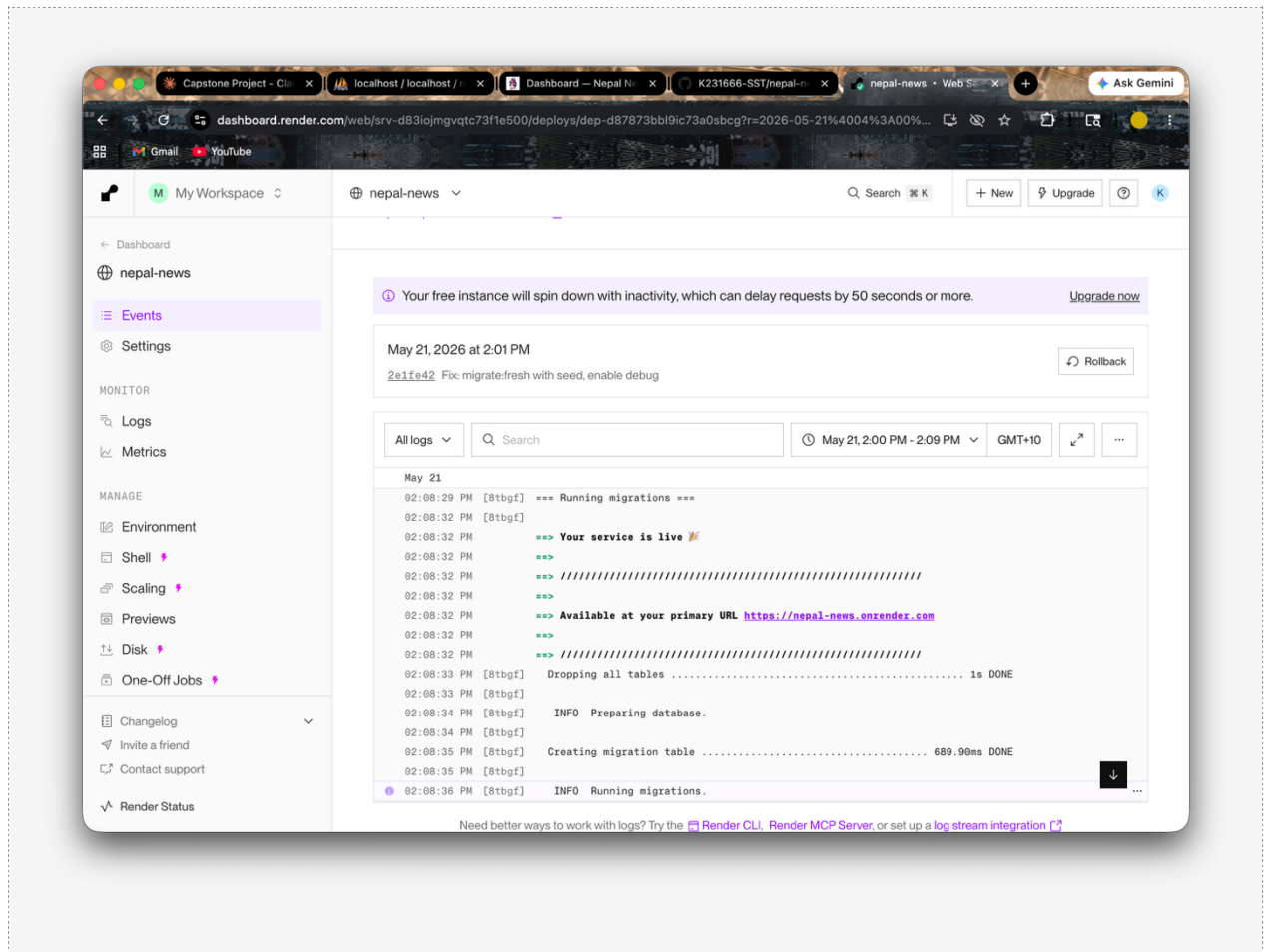
11. Deployment Documentation

11.1 CI/CD Pipeline — GitHub to Render

Nepal News Australia uses an automated deployment pipeline. Every push to the GitHub main branch triggers a fresh Docker build and deployment on Render.com.

Step	Action	Automated?
1	Developer commits and pushes to GitHub main branch	Manual (git push)
2	Render.com webhook detects new commit	Automatic
3	Render pulls latest code from GitHub	Automatic
4	Docker build executes (Dockerfile runs)	Automatic
5	Composer installs PHP dependencies	Automatic (Docker)
6	Container starts — start.sh executes	Automatic
7	Laravel migrations	Automatic (start.sh)
8	Config and routes cached for performance	Automatic (start.sh)
9	Apache starts serving the application	Automatic
10	Render health check confirms service is live	Automatic





11.2 Environment Configuration

All sensitive configuration is stored as environment variables in the Render.com dashboard — never in the codebase:

Variable	Value	Purpose
APP_KEY	base64:QEb/yp+...Q	Laravel application encryption key
APP_ENV	production	Enables production optimisations
APP_DEBUG	false	Hides errors from end users
APP_URL	https://nepal-news.onrender.com	Application URL for link generation
DB_PASSWORD	[secured]	Database password — stored in Render only
DB_SSL_CA	/etc/ssl/certs/ca-certificates.crt	SSL certificate for Aiven connection
GROQ_API_KEY	gsk_[secured]	Guru AI chatbot API key
CACHE_STORE	file	File-based caching (no Redis needed)
SESSION_DRIVER	file	File-based sessions

11.3 Local Development Setup Guide

To run Nepal News Australia locally:

20. Install XAMPP with PHP 8.2+ and MySQL
21. Clone repository: `git clone https://github.com/K231666-SST/nepal-news.git`
22. Install dependencies: `composer install`
23. Copy environment file: `cp .env.example .env`
24. Configure .env: `DB_PORT=3307, DB_DATABASE=nepal_news`
25. Generate app key: `php artisan key:generate`
26. Run migrations: `php artisan migrate`
27. Seed database: `php artisan db:seed`
28. Start server: `php artisan serve --port=8080`
29. Visit: `http://localhost:8080`

12. Help Desk and Support Documentation

12.1 Demo Login Credentials

Role	Email	Password	Access
Administrator	admin@nepalnews.com.au	admin123	Full system — articles, users, ads, events
Editor	editor@nepalnews.com.au	editor123	Publish and edit all articles
Contributor	contributor@nepalnews.com.au	contributor123	Create draft articles only
Reader	reader@nepalnews.com.au	reader123	Read-only — no dashboard access

12.2 Common Issues and Solutions

Issue	Cause	Solution
500 Server Error on homepage	Missing database table or migration not run	Run <code>php artisan migrate:fresh --seed</code>
Login not working	Session driver misconfigured	Check <code>SESSION_DRIVER=file</code> in <code>.env</code>
Guru AI not responding	Groq API key missing or quota exceeded	Check <code>GROQ_API_KEY</code> in environment variables
Weather widget not loading	API key missing	Set <code>OPENWEATHER_API_KEY</code> in <code>.env</code>
Images not showing	Storage link not created	Run <code>php artisan storage:link</code>
CSS not loading	Public assets not found	Ensure <code>public/assets/</code> folder exists and Apache serves <code>public/</code>
Database connection failed	Wrong DB credentials	Verify <code>DB_HOST</code> , <code>DB_PORT</code> , <code>DB_PASSWORD</code> in <code>.env</code>
Admin routes returning 403	User role not set correctly	Check <code>users.role</code> column in database

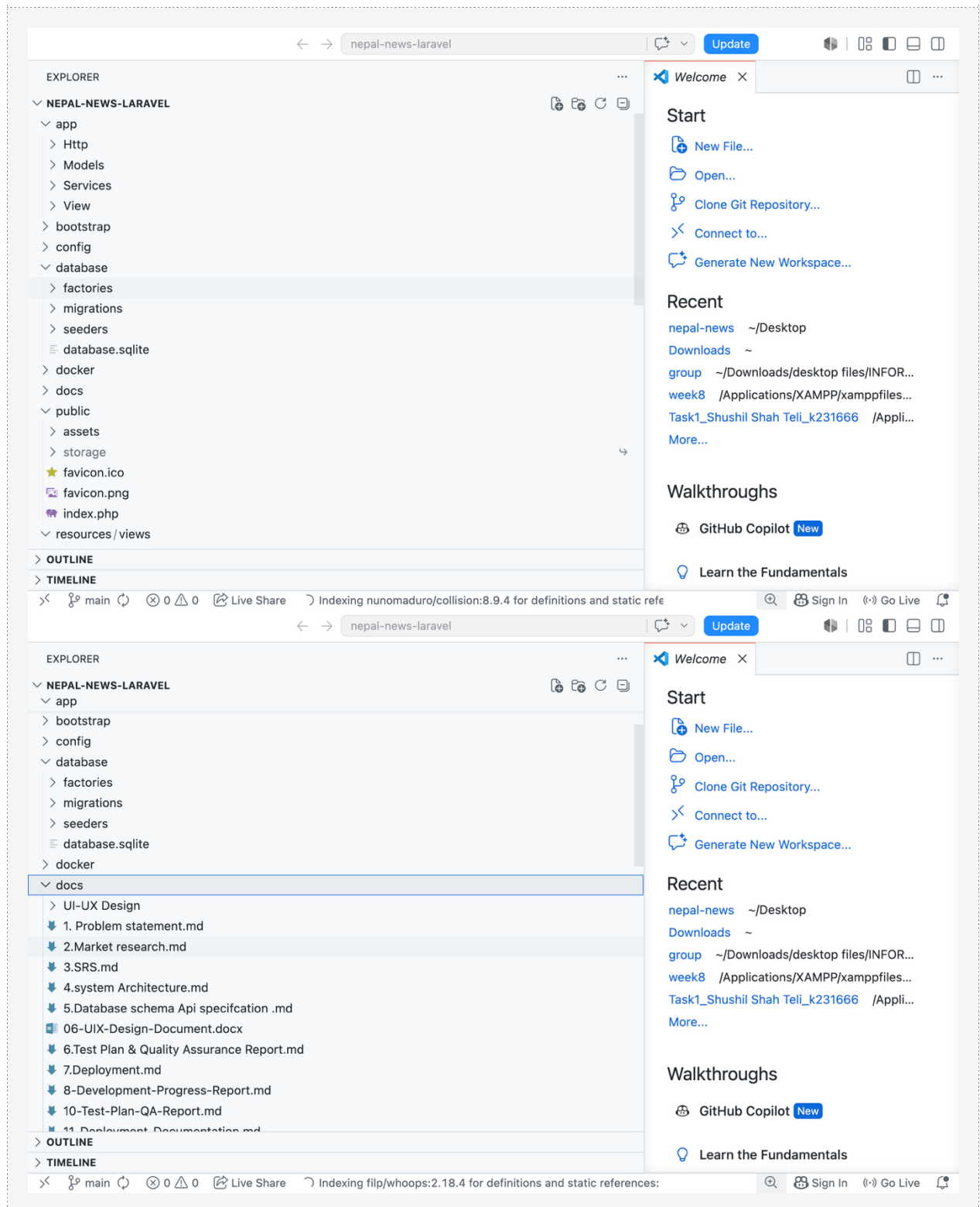
12.3 Maintenance Guide

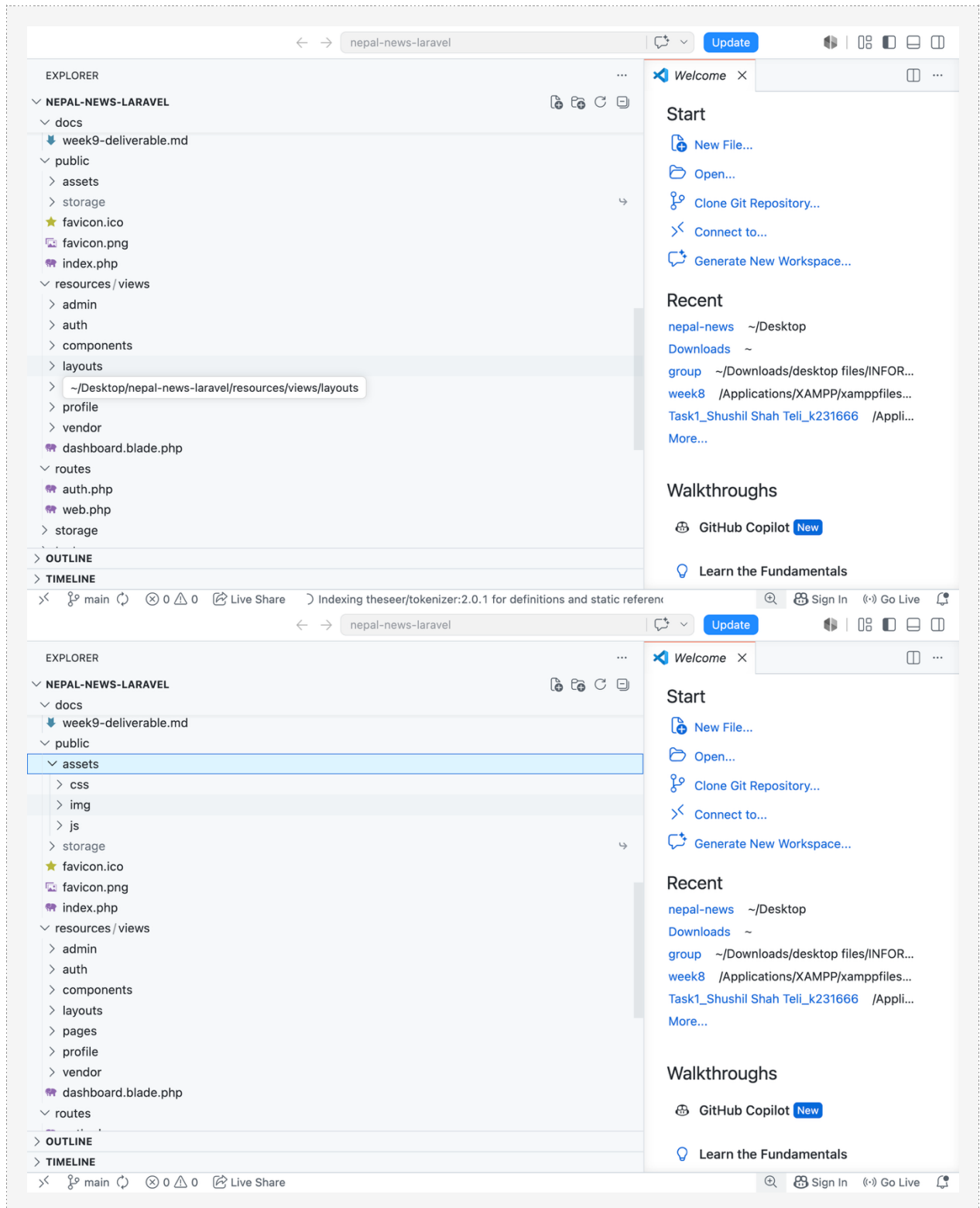
- Backups: Aiven.io provides automatic daily backups on paid plans; export manually via phpMyAdmin for free tier
- Laravel logs: Check storage/logs/laravel.log for application errors
- Render logs: View in Render Dashboard → Logs tab for real-time server errors
- Dependency updates: Run composer update periodically and test locally before deploying
- Database migrations: Any schema changes must have a migration file — never edit schema directly
- API keys: Rotate API keys regularly via each provider's dashboard
- Content moderation: Admin users can archive/delete inappropriate articles via /dashboard

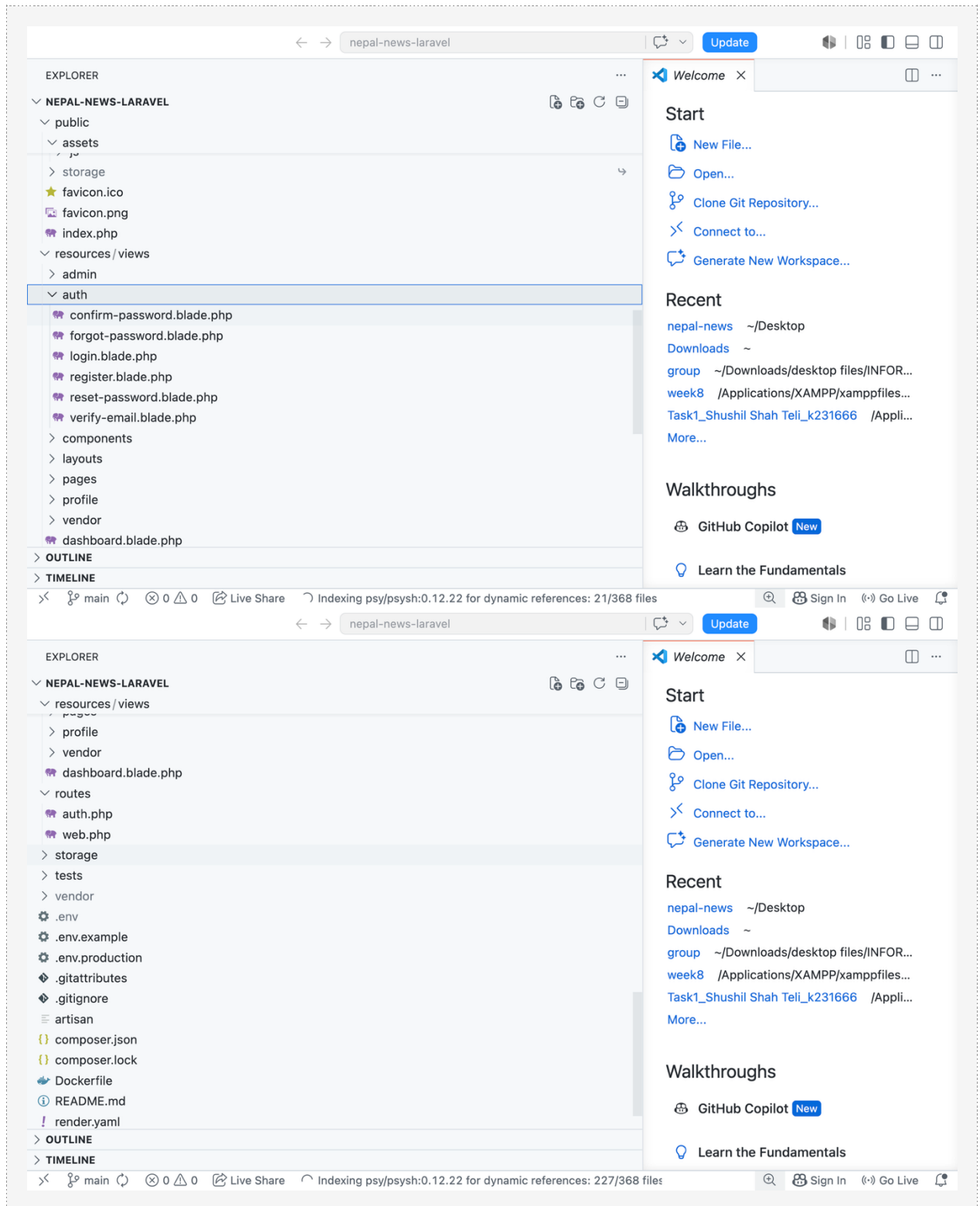
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Appendix A — Project File Structure







Appendix B — GitHub Repository Evidence

The screenshot displays the GitHub repository page for **K231666-SST/nepal-news**. The repository is public and has 44 commits. The commit history is as follows:

Commit	Message	Time
K231666-SST	Fix: skip fulltext index for SQLite compatibility	12 hours ago
	Update Guru AI, CSS fixes, sidebar, routes, all modified files	last week
	Update Guru AI, CSS fixes, sidebar, routes, all modified files	last week
	Update Guru AI, CSS fixes, sidebar, routes, all modified files	last week
	Fix: skip fulltext index for SQLite compatibility	12 hours ago
	Fix: restore port 80, fix apache.conf permissions	2 days ago
	Add Week 8, 9, 10, 11 deliverable documentation	last week
	Update Guru AI, CSS fixes, sidebar, routes, all modified files	last week
	Update Guru AI, CSS fixes, sidebar, routes, all modified files	last week
	Update Guru AI, CSS fixes, sidebar, routes, all modified files	last week
	Nepal News Australia — CPRO306 Capstone Project Team 9	2 weeks ago
	Nepal News Australia — CPRO306 Capstone Project Team 9	2 weeks ago
	Fix 500 error - add env setup in Docker	2 weeks ago
	Fix APP_URL to nepal-news.onrender.com	last week
	Nepal News Australia — CPRO306 Capstone Project Team 9	2 weeks ago
	Switch to SQLite — no more MySQL dependency	13 hours ago
	Remove npm from Dockerfile - no package.json exists	13 hours ago
	Clean repo, add README and author documentation	2 weeks ago
	Nepal News Australia — CPRO306 Capstone Project Team 9	2 weeks ago
	Nepal News Australia — CPRO306 Capstone Project Team 9	2 weeks ago
	Nepal News Australia — CPRO306 Capstone Project Team 9	2 weeks ago
	Switch to SQLite — no more MySQL dependency	13 hours ago
	Switch to SQLite — no more MySQL dependency	13 hours ago

The repository structure includes the following files and folders:

- app
- bootstrap
- config
- database
- docker
- docs
- public
- resources/views
- routes
- storage
- tests
- env.example
- env.production
- gitattributes
- .gitignore
- Dockerfile
- README.md
- artisan
- composer.json
- composer.lock
- render.yaml
- start.sh

The repository also features a README file and a list of contributors. The languages used in the repository are Blade (45.9%), PHP (29.6%), CSS (16.3%), JavaScript (7.6%), and Other (0.7%).

Appendix C — Live Website Screenshots

